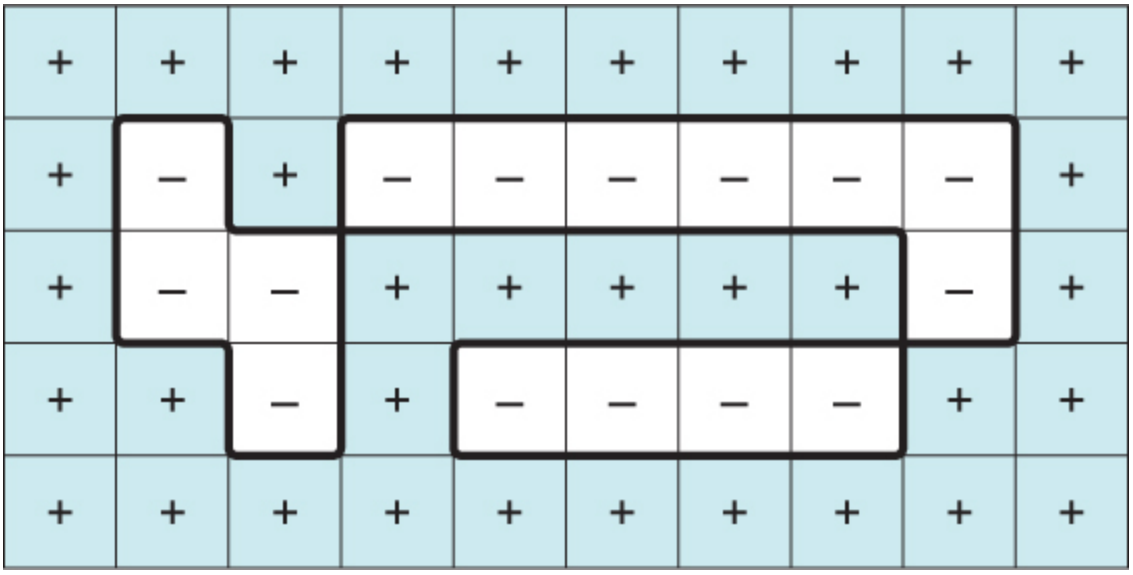
Remarkably, we can model the behavior of many electrons and observe the large-scale emergent organization with multi-agent reinforcement learning. All we need to do is interpret the energy of an electron as its "reward." If an electron changes its spin to align with its neighbor, we will give it a positive reward; if it decides to anti-align, we give it a negative reward. When all the electrons are trying to maximize their rewards, this is the same as trying to minimize their energy, and we will get the same result the physicists get when they use energy-based models.

You might wonder why these modeled electrons wouldn't just all align in the same direction rather than form domains like a real magnet if the electrons get positive rewards for being aligned. Our model is not completely realistic, but it does end up forming domains because with a sufficiently large number of electrons, it becomes increasingly improbable for all of them to align in the same direction, given that there is some randomness to the process (figure 9.11).

Figure 9.11. This is a depiction of a 2D Ising model of electron spins where + is spin-up and – is spin-down. There is a domain of electrons that are all spin-down (highlighted in black), and these are surrounded by a shell of spin-up electrons.



As you'll see, we can also model the temperature of the system by changing the amount of exploration and exploitation. Remember, exploration